SWIMMING POOL SHOCK

What is pool shock? Pool shock refers to a granular oxidizer, which is a powdered form of chlorine used for pool water treatment. In addition to the noun pool shock, it can also be used as a verb, in the act of shocking a pool. When you shock a pool, it means you're adding granular oxidizer (pool shock) to the water in sufficient quantity for the desired change.

WHEN TO USE POOL SHOCK: THE A-B-CS OF SHOCKING POOL WATER



Why do you need to shock a pool? Pools are shocked for a few different reasons. One is to remove combined chlorine molecules, aka chloramines, from the water. Another common reason to shock a pool is to remove excess bather waste and

bacteria after heavy pool use, contamination events, or to treat visible algae in the water. Bromine-treated pools and spas also use pool shock to reactivate bromide ions in the water.

There are really 3 main reasons to shock a pool or spa, raising the Free Available Chlorine level high enough (10–30 ppm) to oxidize or destroy the offending contaminants. We like to call these the ***A-B-Cs of Pool Shock.***

Algae: Green, yellow, pink, or black, the best algaecide is chlorine, and lots of it. Pool algae growth can be controlled with algaecide. But to kill algae and clear the pool, you'll need to use pool shock. A pH on the slightly low side of the ideal range, or around 7.2–7.4, will allow the chlorine shock to be most potent. Some types of pool shock, including cal-hypo and liquid chlorine, will raise the pH slightly. Stabilized dichlor shock has a near-neutral pH. Depending on the severity of the algae, 10–30 ppm chlorination is needed to kill active algae blooms. Chlorine accelerators, such as Yellow Out work to boost your chlorine level to fight severe algae blooms of all colors.

Bacteria & Bather Waste: Bacteria can enter the pool from many sources, and most of it is harmless. However, pathogenic bacteria may also exist. Use chlorine shock to remove bacteria after extensive pool use, severe storms, heavy rainfall, long winters, or to address swimmer "accidents." Bather waste includes any contaminants brought into the pool by swimmers, including dead skin cells, hair, lotions, cosmetics, and soaps, as well as more potent contaminants like sweat, fungus, and yes, even urine and feces.

Chloramines, Contaminants, Cloudy Water: Combined chlorine molecules are responsible for "red eye" swimmers and a strong chlorine odor. When chloramine levels exceed 0.3 ppm (Total Chlorine - Free Chlorine = Combined Chlorine), add enough chlorine or non-chlorine shock to break apart the combined chlorine molecules. This is usually around 10–20x the tested CC level. As mentioned in the previous paragraph, contaminants in the water are a reason to shock the water so the pool stays sanitized. And last, but not least, cloudy pool water is often a symptom of poor water balance and low sanitizer levels. Balance the water and shock the pool to get the water crystal clear again.

OTHER TIMES TO USE POOL SHOCK

Shocking the pool helps break down all sorts of organic contaminants in your pool water. Lots of harmless matter gets into a pool on a daily basis, which can negatively

impact water balance and sanitization. Cloudy water, often caused by these extra microscopic contaminants, can usually be corrected by superchlorinating the pool or adding non-chlorine oxidizer to destroy the suspended particles that cause the hazy water conditions.

In short, for best results, use chlorine shock for algae and bacteria removal, and use non-chlorine shock for the removal of non-living organic contaminants and chloramines. Also, if your Free Available Chlorine level is less than 2.0 ppm, use a chlorine shock to bring the levels up and keep the pool sanitized.

Start-Up and Close-Down: For winterized pools, opening the pool is the first time of the year that pool shock is used, after the water chemistry is balanced. This helps oxidize particles, kill bacteria and algae, and restore water clarity after a long winter under cover. Prior to closing the pool, pool shock is used to disinfect the water in preparation for the long winter months ahead.

Heavy Rain: Rain is pure water, distilled by evaporation. But as it falls through the air, it picks up airborne particles that then end up in your pool. Air pollution, dust, pollen, and algae spores can discolor the water, consume your chlorine, and affect water chemistry. During heavy storms, overhanging trees, overflowing planters, or lawn areas next to the pool can wash in soil laden with bacteria and phosphates, in addition to plenty of tree and plant debris, which can quickly use up your chlorine residual.

Chloramines: We mentioned this one above in our A-B-Cs of pool shock, but it bears repeating because it's such a common issue. Free Available Chlorine becomes Combined Chlorine when it bonds to nitrogen or ammonia in the water. This bond renders the chlorine molecule useless, and causes the pool water to smell strongly of chlorine. These chloramines can irritate swimmers' eyes and skin, and make for a generally unpleasant swimming experience. DPD test kits can test for both Free and Total Chlorine, with any difference in the test results indicating your pool's levels of Combined Chlorine. Again, shock the pool to remove chloramines when levels exceed 0.3 ppm.

HOW TO SHOCK A POOL

1. Check pH levels and adjust as needed to a low range of 7.2–7.4. This will allow your chlorine to work most efficiently.
2. Clean the pool to remove leaves and debris. Also remove the pool cleaner.

3. Determine how much shock is needed to fix the situation.
4. Add shock directly to the pool, according to product label instructions. In most scenarios, vinyl pools should pre-dissolve chlorine shock to avoid bleaching and discoloration of the liner.
5. Add shock when the sun is not shining directly on the pool. Nighttime shocking is usually best.
6. Run the pump and brush the walls and floor to help distribute and circulate the shock throughout your pool.
7. Check chlorine levels before swimming. For most chlorine shocks, you'll need to wait at least 12–24 hours.

HOW OFTEN TO USE POOL SHOCK



How often do you need to shock a pool? Every pool is different, but we strongly suggest shocking the pool at least once per week. Most pools should be routinely shocked to remove bacteria, algae, chloramines or other contaminants, or to help clear cloudy pool water or some other water problem. You can test for chloramines, and you can see algae. However, bacteria and other disease-causing pathogens are invisible, as are nitrates, phosphates, and ammonia. For this reason, most pool owners shock the pool weekly, even if the water appears clean and clear. This will help ensure it's properly disinfected and safe for swimming. You don't have to use chlorine shock every week, though! When your Free Available Chlorine levels are 2.0 ppm or higher, you can use chlorine-free shock to oxidize the pool and control chloramines. When used regularly, chlorine-free shock can help maximize chlorine efficiency in your pool. The best part is that you can swim almost immediately after using non-chlorine pool shock — no need to wait for hours!

For pools that have hosted a party for a dozen or more swimmers, or if a pool becomes contaminated with urine, feces, or vomit, a strong shocking with chlorine shock is in order. In cases where chlorine levels have depleted, due to hot temperatures, faulty equipment, or operator error, chlorine pool shock can be used to quickly raise chlorine levels.

Another reason to use pool shock more frequently is when a pool cannot obtain a good chlorine reading. If the water is balanced, algae-free, and does not have excessive levels of Cyanuric Acid, but still cannot get a good chlorine reading despite continuous additions of chlorine, there may be high levels of nitrates or ammonia consuming the chlorine. In such cases, where you have tried everything else, but the chlorine demand seems insatiable, try "triple shocking" the pool. Use 3 pounds of shock per 10,000 gallons of pool water to destroy these invisible consumers of chlorine and get your chlorine demand back under control. Just remember to wait until the Free Available Chlorine level has returned to 2.0–4.0 ppm before you hop back in the pool again. Once you have the sanitizer levels under control, you can resume a routine weekly shocking cadence.

CHEMICALS USED AS POOL SHOCK

For a pool that uses chlorine or bromine for daily chlorination, there are a variety of EPA-approved chemicals that can be used to shock the pool. The best one for your pool may depend on your pool type, or if you have issues with hard water or high Cyanuric Acid levels.

Calcium Hypochlorite: Cal-hypo for short, is the most economical pool shock you can buy. It is available in 65% and 73% strength, has a pH level of 12, and is not stabilized.

Sodium Dichlor: Dichlor for short, is stabilized pool shock, meaning it contains Cyanuric Acid to protect your chlorine against degradation from the sun. It's available in 56% strength, with a nearly neutral pH level. Unlike cal-hypo, it does not add any calcium to the pool.

Potassium Monopersulfate: Non-chlorine shock oxidizes pool water in a chlorine-free formula. It's not affected by sunlight, leaves no residue, and you can swim as soon as 15 minutes after application. While it won't increase chlorine levels in your water, it will help the existing chlorine residual work as efficiently as possible.

Which pool shock to use? Chlorine shock treatments come in two types — ***cal-hypo and dichlor***. Calcium hypochlorite is the most common, strongest, and cheapest pool shock. Dichlor is a stabilized granular pool shock, made with stabilizer to protect it from the sun and keep it active longer during the day.

Most pools can use cal-hypo to avoid adding extra Cyanuric Acid to the water, especially when they're already using stabilized trichlor tablets. But for those in hard water areas, with concerns of growing Calcium Hardness levels, sodium dichlor along with non-chlorine shock may be a better choice. Each pound of cal-hypo will add 5-7 ppm to Calcium Hardness levels. Vinyl pools benefit from non-chlorine shock, which won't bleach or fade vinyl liners, and also don't cloud the water or leave behind a dusty residue. Pools with growing Cyanuric Acid levels may want to avoid dichlor shock, as each pound adds a small amount of Cyanuric Acid.

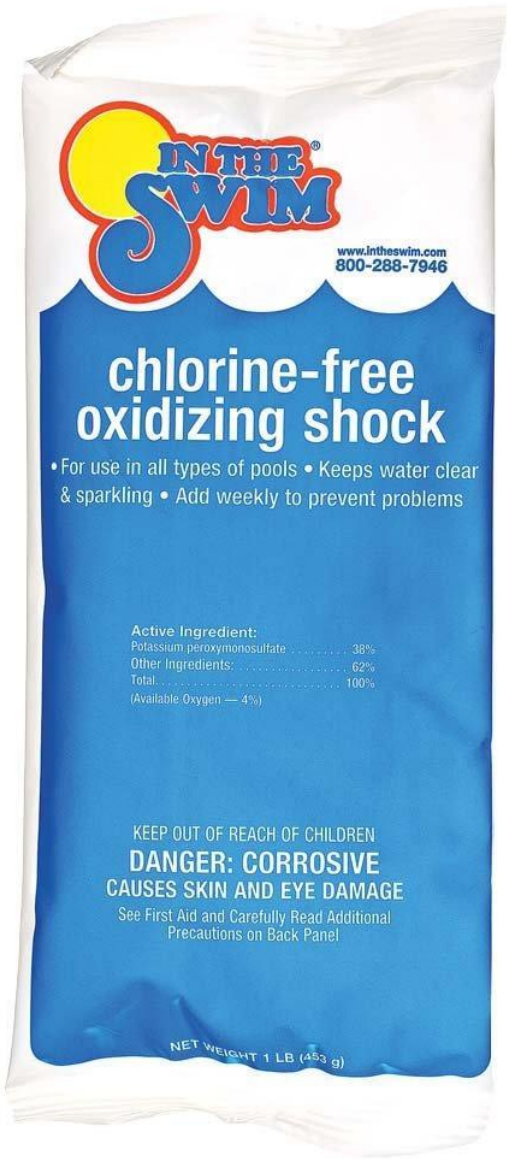
In The Swim cal-hypo pool shock is packaged in easy-opening 1-pound bags, in our 65% Pool Shock. We also have Instant Pool Shock in 1.07-pound bottles, and cal-hypo granules in 25- and 50-pound buckets, which can make it easier to use when adding large amounts of pool shock at one time.

Chlorine-free shock treatments contain oxygen with salts of potassium, and it has many advantages compared to chlorine-based pool shock. In addition, it costs the same. Strengths vary among brands, but most non-chlorine shock is 38–42% potassium monopersulfate, *aka* MPS.

Chlorine-free, cal-hypo, and dichlor pool shocks are all available in 1-pound bags.

One last question we get asked about the different types of pool shock: How long should you wait to swim after shocking the pool? Well, it depends on which type of shock you use! In all cases, carefully follow the product label instructions, which will guide you in how long to wait after shocking before swimming. Heavy shocking with granular chlorine will generally require 24–48 hours before the chlorine level has dropped to safe swimming levels (below 5 ppm). Lithium and non-chlorine shock labels typically allow immediate swimming or a brief 15-minute waiting period, but check the package label to be sure.

PROS & CONS OF NON-CHLORINE POOL SHOCK



PROS:

- Quick-release formula, fast acting, no residue, no odor
- Near-neutral pH
- Oxidizes organic contaminants to purify water
- Removes chloramines and ammonia
- No pre-dissolving needed — just pour into pool
- You can usually swim immediately; no waiting period like with chlorine shock
- Won't bleach or fade vinyl liners or swim suits
- Won't add Calcium Hardness or Cyanuric Acid to pool

CONS:

- Not effective for algae treatment
- Not effective for bacteria treatment
- Only recommended for use if residual Free Available Chlorine levels are at 2.0 ppm or above

PROS & CONS OF CHLORINE POOL SHOCK



PROS:

- Quick-release, fast-acting formula
- Oxidizes both living and non-living organic contaminants to purify water
- Removes chloramines and ammonia
- Kills algae and bacteria quickly to sanitize the water

CONS:

- In some cases, pre-dissolving granular is recommended for surface protection — always read and follow product label instructions
- Waiting period of 12—24 hours is generally recommended before swimming again
- High levels of chlorine can be harsh on soft and shiny surfaces
- Cal-hypo increases Calcium Hardness, and dichlor adds Cyanuric Acid (this can be considered a "pro" in certain scenarios)
- More hazardous to store and use, and can have a strong odor
- Residue of "shock dust" can be left behind, or water may become cloudy as the shock fights off contaminants

ESTIMATING POOL SHOCK AMOUNT: HOW MUCH SHOCK SHOULD I USE?

Wondering how much pool shock to use? You'll need to first know the amount of water in your pool, give or take a few hundred gallons. If you aren't sure, take some measurements and consult an online pool volume calculator. Generally speaking, the dosage amount of pool shock is 1 pound per 10,000 gallons, but always consult the shock package label for precise instructions. Depending on the severity of the problem you are addressing, you may need a double or triple dose to successfully fix the issue.

For chloramine removal, shock the pool to reach a Free Chlorine level 10–20 times the amount of measured chloramines. For algae removal, 30 ppm is a generally accepted target. However, you may use more or less depending on the severity of the algae bloom. Another rule of thumb in algae removal is that if the water is still green, you need more. For boosting chlorine levels that have dropped to zero, 1 pound of pool shock will usually be sufficient.

In addition, knowing how much pool shock you should use depends on what problem you are trying to solve and the current condition of the water. If the water looks good and you simply want to boost the chlorine level a bit, add 1 bag per 20,000 gallons. If the water is hazy or cloudy, use a full bag per 10,000 gallons. Or, if there is algae, use 2–6 lbs per 10,000 gallons, depending on the extent of the algae bloom. A chlorine level between 10 and 30 ppm, sustained for several hours, is usually needed to remove severe algae, bacteria, and chloramines. Super Pool Shock can deliver 10 ppm per 10,000 gals from a 1-pound bag in good water conditions. If water conditions are poor, you may need more.

In many cases, the dosage listed on a bag of shock will be effective on blue and clear water, but if you have very cloudy or very green water, a 3–6x treatment dosage is not unusual. The higher the level of contaminants, algae, or chloramines in the water, the more pool shock will be needed to oxidize and break down the matter. Visibility (or lack thereof) is another way to gauge the severity of an algae bloom. If you can see the shallow end floor, use a double dose of shock. But if you can only see down for 12" or 24" into the water, use a 3–6x treatment dosage.

Remember that a low pH level is crucial to successfully shocking a pool. At a pH level of 8.0, over half of your shock is ineffective and wasted. At a pH level of 7.2, however, over 90% of your shock will become active algae and bacteria killers.

Missing the mark sometimes happens when trying to clear adverse water conditions. If you still have a strong chlorine level 12 hours after shocking, and the water appearance is improving with filtering, consider it mission accomplished (more than likely). But if the chlorine level is zero again after 12 hours, and the pool doesn't look much better, you may have missed the threshold of breakpoint chlorination. Try again!

POOL SHOCK SHELF LIFE



How long does pool shock last? Granular chlorine products will lose only a small percentage of potency when stored in a cool, dry, and dark location. When stored in a shed or garage, however, the varying temperature and humidity levels will begin to solidify the contents, and within a few years, the 1-pound plastic bags will deteriorate.

For longer and safer storage, we recommend buying loose cal-hypo in buckets, and/or non-chlorine shock. Store in a dark and cool location with a very tight lid to keep out moisture and contamination, as well as to prevent off-gassing.

PRE-DISSOLVING POOL SHOCK IN A BUCKET

For vinyl liner pools, undissolved chlorine granules resting directly on vinyl can bleach, fade, or corrode soft vinyl surfaces. Pre-dissolving is accomplished by filling a clean 5-gallon bucket full of pool water. Whenever pre-dissolving pool chemicals, always refer to the product label for detailed instructions, as sometimes the recommended gallon-to-granule ratio will differ between chemicals.

Next, pour the correct amount of granular pool shock directly into the water (always add chemicals to water, not water to chemicals). Stir with a yardstick or suitable paddle for several minutes to dissolve the granules.

Finally, pour the solution around the edge of the pool. When the bucket is almost empty, stop pouring it into the pool, and add more water to finish dissolving any remaining granules in the bottom of the bucket. Vinyl pools can avoid pre-dissolving when using non-chlorine shock or liquid chlorine.

POOL SHOCKING TIPS

- Balance pH to the low end of the ideal pH range (around 7.2–7.4) before shocking. This will give your shock the most powerful effect.
- Add pool shock on its own, as it can destroy or disrupt other treatment chemicals.
- Never allow pool shock to become hot, moist, or contaminated with dirt and debris.
- Never allow pool shock to mix with any other pool chemical, even the same type.
- Never store opened bags of shock, which can spill. Use the entire bag at one time.
- Never pour pool shock into the skimmer.
- For vinyl liner pools, it's recommended to pre-dissolve your chlorine shock before adding to the water.
- When broadcasting shock across the surface, be mindful of the wind direction. Pour chlorine close to the water's surface.
- Brush the pool after shocking, and filter the water for at least 8 hours afterwards.
- If chlorine level is zero within 8 hours of shocking pool, shock the pool again with even more chlorine.
- Shock your pool after the sun goes down to reduce the effects of UV degradation.

WHEN AND WHY TO SHOCK A POOL

1. To reduce the buildup of micro-contaminants and organic debris.
2. To destroy both harmless and pathogenic bacteria in the water.
3. To remove combined chlorine molecules, aka chloramines.
4. To kill algae.
5. To treat cloudy pool water.
6. To address severe contamination incidents in the pool, such as after a storm or fecal incident.

OTHER USES FOR POOL SHOCK

- Organic stain removal on a plaster pool surface
- Raising chlorine levels fast when it's discovered to be low or near zero
- Raising or maintaining chlorine levels during pump, filter, or salt system troubles
- Spot treating small algae colonies in corners and crevices

POOL SHOCK SAFETY

Storing Pool Shock: Keep in a cool, dry area, separated from other pool chemicals, and out of reach of children. If you purchase shock bags instead of buckets, it's more safely stored if removed from the cardboard box and placed inside a clean bucket or storage bin with a tight-fitting lid. Do not store half-used bags of shock, which can easily spill, become contaminated, or take on moisture. All of these scenarios are hazardous. It's best to always use the entire 1-pound bag at one time. For bucketed shock, keep the lid tightly sealed, and store it in a cool, dry area according to label instructions.

Using Pool Shock: Cut the bag carefully with scissors, and pour it into the water while walking along the pool edge. Use a pool brush to distribute, and sweep or wash any spills into the pool. Vinyl liner pools should pre-dissolve granular shock, unless using quick-dissolving chlorine-free shock.

Never Mix Pool Shock: With anything except water. Pool shock is very reactive. When mixed with any substance other than water, it can release toxic gases, ignite, or even explode. Never place shock into a chlorinator or floater, nor should you ever add it to the skimmer. Always add shock directly to the pool, according to the directions on the product label.

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